

01. Introduction

Lung cancer is one of the leading causes of cancer-related deaths worldwide. It develops when abnormal cells in the lungs grow uncontrollably, forming tumors that interfere with normal lung function.

Early diagnosis significantly improves treatment success and survival rates.

02. What is Lung Cancer?

Lung cancer begins in the tissues of the lungs, usually in the cells lining the airways (bronchi).

There are two main types:

Non-Small Cell Lung Cancer (NSCLC)

- Most common type
- Accounts for about 85% of cases
- Usually grows more slowly

Small Cell Lung Cancer (SCLC)

- Less common
 - Grows rapidly
 - Spreads quickly to other organs
 - Often associated with heavy smoking
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03. Risk Factors

Several factors increase the risk of lung cancer.

Major Risk Factors

- Cigarette smoking
- Passive (secondhand) smoking
- Exposure to radon gas
- Air pollution
- Asbestos exposure
- Family history of lung cancer
- Previous radiation therapy to the chest
- Occupational exposure to harmful chemicals

Smoking remains the leading cause of lung cancer.

04. Common Symptoms

Early lung cancer may not produce noticeable symptoms.

As the disease progresses, symptoms may include:

- Persistent cough
- Coughing up blood
- Chest pain
- Shortness of breath
- Wheezing
- Hoarseness
- Unexplained weight loss
- Fatigue
- Frequent lung infections
- Loss of appetite

Patients should seek medical evaluation if symptoms persist.

05. Diagnosis

Diagnosis involves several investigations.

Medical History

The doctor asks about:

- Smoking history
- Occupational exposure
- Symptoms
- Family history

Physical Examination

Includes examination of:

- Chest
 - Lymph nodes
 - Breathing sounds
 - General health
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06. Chest X-Ray

A chest X-ray is often the first imaging test.

It may reveal:

- Lung masses
 - Nodules
 - Fluid around the lungs
 - Enlarged lymph nodes
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07. CT Scan

Computed Tomography (CT) provides detailed cross-sectional images of the lungs.

It helps determine:

- Tumor size
- Tumor location
- Spread to nearby structures

CT scans are essential for staging.

08. PET Scan

PET scans detect areas of increased metabolic activity.

They help identify:

- Cancer spread
 - Lymph node involvement
 - Distant metastasis
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09. Bronchoscopy

A bronchoscope is inserted through the mouth or nose into the lungs.

It allows doctors to:

- View the airways
 - Collect tissue samples
 - Remove small obstructions
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10. Lung Biopsy

A biopsy confirms the diagnosis.

Methods include:

- Needle biopsy
- Bronchoscopic biopsy
- Surgical biopsy

The tissue is examined under a microscope.

11. Lung Cancer Staging

Stages describe how far cancer has spread.

Stage I

Cancer remains confined to the lung.

Stage II

Cancer has spread to nearby lymph nodes.

Stage III

Cancer involves nearby organs or multiple lymph nodes.

Stage IV

Cancer has spread to distant organs such as:

- Brain
 - Liver
 - Bones
 - Adrenal glands
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12. Treatment Options

Treatment depends on:

- Cancer type
 - Stage
 - Overall health
 - Lung function
 - Patient preference
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13. Surgery

Surgery is suitable for early-stage NSCLC.

Common procedures include:

Wedge Resection

Removes a small part of the lung.

Lobectomy

Removes one lobe of the lung.

Pneumonectomy

Removes an entire lung.

Surgery aims to completely remove the tumor.

14. Chemotherapy

Chemotherapy uses medicines to destroy cancer cells.

It may be given:

- **Before surgery**
- **After surgery**
- **Along with radiation**
- **For advanced disease**

Common side effects include:

- **Hair loss**
 - **Fatigue**
 - **Nausea**
 - **Infection risk**
 - **Low blood counts**
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15. Radiation Therapy

Radiation therapy uses high-energy rays to kill cancer cells.

It is commonly used:

- **After surgery**
 - **When surgery is not possible**
 - **To relieve symptoms in advanced disease**
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16. Targeted Therapy

Some lung cancers have specific genetic mutations.

Targeted medicines may attack:

- EGFR mutations
- ALK rearrangements
- ROS1 mutations

Targeted therapy often causes fewer side effects than chemotherapy.

17. Immunotherapy

Immunotherapy strengthens the immune system to attack cancer cells.

It is widely used in advanced non-small cell lung cancer.

Benefits include:

- Improved survival
 - Better disease control
 - Longer-lasting responses in selected patients
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18. Possible Complications

Lung cancer may cause:

- Breathing difficulty
- Lung collapse
- Bleeding
- Pneumonia
- Pleural effusion
- Bone pain
- Brain metastasis

Early treatment reduces these complications.

19. Prevention

Many lung cancer cases can be prevented.

Recommendations include:

- Do not smoke.
- Quit smoking immediately.

- Avoid secondhand smoke.
 - Test homes for radon.
 - Wear protective equipment at work.
 - Reduce exposure to air pollution.
 - Maintain regular exercise.
 - Eat a healthy diet.
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20. Living with Lung Cancer

Patients can improve quality of life by:

- Taking medications regularly
 - Following treatment schedules
 - Eating nutritious meals
 - Staying physically active
 - Participating in pulmonary rehabilitation
 - Seeking emotional support
 - Joining cancer support groups
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21. Follow-Up Care

Regular follow-up appointments help:

- Detect recurrence
- Monitor treatment response
- Manage side effects
- Improve long-term survival

Patients should report:

- Persistent cough
 - Chest pain
 - New breathing difficulty
 - Unexplained weight loss
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22. Key Points

- Lung cancer is one of the most common causes of cancer-related deaths.

- Smoking is the biggest risk factor.
- Persistent cough and coughing blood require medical evaluation.
- CT scan and biopsy confirm diagnosis.
- Treatment includes surgery, chemotherapy, radiation therapy, targeted therapy, and immunotherapy.
- Quitting smoking greatly reduces lung cancer risk.
- Early diagnosis significantly improves survival and treatment success.